

The D-Score Snapshot Alignment Theorem

A delay-variance budget for asynchronous biodiversity indicators

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Abstract

Biodiversity composites often fuse observations collected on incompatible clocks: satellite products may refresh daily, field inventories seasonally, and genetic surveys over years. A weighted average can therefore appear precise while its components describe different ecological states. We isolate this temporal-coherence problem in a local common-mode model. For normalized nonnegative effective weights w_i , observation delays τ_i , and harmonic frequency ω , the fused response has alignment efficiency $\eta(\omega) = |\sum_i w_i e^{-i\omega\tau_i}|^2$. We derive the exact pairwise loss identity and the conservative bound $\eta(\omega) \geq 1 - \omega^2 \text{Var}_w(\tau)$. Consequently, a declared ecological bandwidth Ω and target coherence q imply an auditable sufficient rule, $\text{Var}_w(\tau) \leq (1 - q)/\Omega^2$. Equal delays preserve perfect coherence, whereas two equally weighted channels separated by half a cycle cancel completely. A preregistered standard-library Python gate passed 21 of 21 analytic, randomized, convergence, negative-control, and time-domain checks. Separately, Aristotle returned six Lean theorems that an independent local audit rebuilt in Lean 4.28.0/Mathlib, with no proof holes, forbidden constructs, or axioms beyond `propext`, `Classical.choice`, and `Quot.sound`. The result is a formally verified conditional signal-model statement, not an empirical validation of D-Score, a novelty or priority finding, or evidence of field utility.

Keywords: biodiversity monitoring; data fusion; temporal alignment; composite indicators; D-Score; sensor synchronization; measurement coherence

1 Introduction

Composite biodiversity indicators are assembled from measurements that differ in biological meaning, spatial support, error structure, sampling protocol, and time. The last difference is easy to hide. A species inventory dated 2021, a satellite productivity layer dated 2026, and a genetic-diversity estimate dated 2018 can be placed in the same row of a database and averaged without any arithmetic warning. The result is numerically well formed but may not describe an ecosystem that ever existed at one time.

The D-Score research program makes this concern concrete. Its validation protocol combines taxonomic, phylogenetic, genetic, and environmental inputs. Prior Viridis work has separately studied component priorities, finite-sample measurement floors, spatial allocation of monitoring power, optimal revisit cadence, and weights robust to strategic gaming. None of those results answers the narrower question here: *when do component timestamps permit a weighted score to be interpreted as a coherent snapshot?*

The broader literature supplies both motivation and caution. Multi-sensor fusion systems require a shared time reference and explicit temporal sample alignment; experimental work shows that software synchronization can prevent cumulative desynchronization across heterogeneous sensors [1]. Biodiversity data products similarly combine remote sensing, field surveys, and other observation streams, for which fusion can add substantial information [5]. Yet ecological timing is not merely an instrumentation nuisance. Global phenology maps reveal spatially displaced seasonal cycles and validate them against independent products [2]; nearly half a million plant and animal time series show divergent phenological shifts [3]. Real biological asynchrony can be signal rather than error. Monitoring uncertainty also limits the detectable evidence for biodiversity trend change even at large sample sizes [4].

We therefore seek a deliberately bounded result. We do not claim that all D-Score components share one dynamic, that all delays should be removed, or that temporal coherence proves ecological validity. We ask what follows *if* a set of normalized channels is intended to measure a locally common dynamical mode. Under that contract, delay becomes phase, the weighted snapshot becomes a phasor sum, and an exact coherence audit follows.

2 Preregistration and scope

The hypotheses, assumptions, falsifiers, and verification program were fixed before the manuscript was drafted.

2.1 Hypotheses

For a locally common harmonic mode and known channel delays:

1. the aligned signal fraction is the squared magnitude of the weighted delay phasor;
2. its loss has an exact pairwise cosine decomposition;
3. the loss is bounded by frequency squared times weighted delay variance;
4. this bound is second-order sharp as delay spread tends to zero; and
5. a bandlimited common-mode process inherits a spectral version of the same guarantee.

2.2 Assumptions

The channels share a locally common mode after centering and scaling. Effective weights are nonnegative and normalized, delays are known relative to a declared reference time, and the local response is linear. Measurement noise, spatial mismatch, missingness, taxonomic bias, calibration drift, and channel-specific ecological dynamics are held outside the timing term. These exclusions are substantive: a real deployment must test them rather than inherit them.

2.3 Falsifiers and controls

The computational gate would fail if the pairwise identity exceeded numerical tolerance, if η left $[0, 1]$, if the variance bound were violated, or if independent time-domain estimates failed to converge to the phasor prediction. Negative controls included equal delays, a static mode, one active channel, balanced half-period cancellation, and full-period phase wrapping.

3 Model

Let the centered common ecological perturbation be

$$x(t) = \Re\{ze^{i\omega t}\}, \quad (1)$$

where z is a complex amplitude and ω is angular frequency. Channel i observes the same local mode at delay τ_i . Channel sensitivity can be absorbed into an effective weight w_i after sign and scale have been validated. We assume

$$w_i \geq 0, \quad \sum_{i=1}^n w_i = 1. \quad (2)$$

The fused response is

$$y(t) = \sum_i w_i x(t - \tau_i) = \Re\{ze^{i\omega t} A(\omega)\}, \quad (3)$$

with transfer phasor

$$A(\omega) = \sum_i w_i e^{-i\omega\tau_i}. \quad (4)$$

We define snapshot alignment efficiency as the common-mode output-to-input energy ratio

$$\eta(\omega) = |A(\omega)|^2. \quad (5)$$

This quantity measures temporal coherence only. It is not an information content, accuracy, causality, or conservation-value score.

Define the weighted mean delay and variance by

$$\bar{\tau} = \sum_i w_i \tau_i, \quad \text{Var}_w(\tau) = \sum_i w_i (\tau_i - \bar{\tau})^2. \quad (6)$$

Only relative timing should matter: translating every τ_i by the same constant rotates A without changing η .

4 Results

The formal artifact `SnapshotAlignment.lean` defines the weighted complex `phasor`, its squared-norm `alignment`, the `weightedMean`, and `delayVariance`. It contains exactly six audited theorems: `phasor_energy_pairwise_identity`, `alignment_efficiency_unit_interval`, `delay_variance_bounds_alignment_loss`, `alignment_budget_sufficient_for_band`, `equal_delay_perfect_alignment`, and `two_channel_half_period_cancellation`. The first is stated over an arbitrary finite index type without weight hypotheses. The next three use nonnegative normalized weights; the band theorem additionally assumes $0 < \Omega$, $0 \leq q \leq 1$, $|\omega| \leq \Omega$, and the displayed variance budget. The equal-delay theorem assumes a nonempty finite channel type and weights summing to one. The cancellation theorem assumes exactly two equal weights and the explicit antipodal-phase equation. These are mathematical claims about the defined model only.

4.1 Exact pairwise identity

Proposition 1 (snapshot alignment identity). Under Eqs. (2)–(5),

$$\eta(\omega) = \sum_i w_i^2 + 2 \sum_{i < j} w_i w_j \cos(\omega(\tau_i - \tau_j)), \quad (7)$$

$$1 - \eta(\omega) = 2 \sum_{i < j} w_i w_j [1 - \cos(\omega(\tau_i - \tau_j))]. \quad (8)$$

Derivation. Expand $A\bar{A}$ into diagonal and off-diagonal terms. Pair each complex exponential with its conjugate to obtain a cosine. Then subtract the result from $1 = (\sum_i w_i)^2 = \sum_i w_i^2 + 2 \sum_{i < j} w_i w_j$.

Equation (8) assigns every unit of modeled coherence loss to a pair of channels. It also makes the bounds $0 \leq \eta \leq 1$ immediate: squared magnitude gives the lower bound and each term in the loss is nonnegative.

4.2 Delay-variance bound

Theorem 1 (delay variance bounds alignment loss). For every real ω ,

$$1 - \eta(\omega) \leq \omega^2 \text{Var}_w(\tau), \quad \eta(\omega) \geq 1 - \omega^2 \text{Var}_w(\tau). \quad (9)$$

Derivation. Apply $1 - \cos u \leq u^2/2$ termwise in Eq. (8), then use the weighted pairwise-variance identity

$$\sum_{i < j} w_i w_j (\tau_i - \tau_j)^2 = \text{Var}_w(\tau). \quad (10)$$

For small delays, $1 - \cos u = u^2/2 + O(u^4)$, so the ratio $(1 - \eta)/(\omega^2 \text{Var}_w)$ tends to one whenever the delay shape has nonzero variance. The gate recovered second-order convergence under repeated halving.

4.3 Bandlimited process and operational rule

The following spectral extension is an ordinary analytic consequence and was not formalized among the six Lean theorems. Suppose a wide-sense stationary common-mode process has spectral density $S_x(\omega)$ supported on $|\omega| \leq \Omega$. By linear filtering and Parseval's relation, its normalized fused common-mode energy is the S_x -weighted average of $\eta(\omega)$. Equation (9) therefore yields

$$\frac{\int S_x(\omega) \eta(\omega) d\omega}{\int S_x(\omega) d\omega} \geq 1 - \Omega^2 \text{Var}_w(\tau). \quad (11)$$

Corollary 1 (alignment budget). If a monitoring product declares a maximum modeled frequency Ω and minimum acceptable alignment $q \in [0, 1]$, then

$$\boxed{\text{Var}_w(\tau) \leq \frac{1 - q}{\Omega^2}} \quad (12)$$

is sufficient for $\eta(\omega) \geq q$ throughout the declared band.

This rule is conservative. It can be audited from timestamps and weights, but it is only meaningful if Ω is scientifically justified. Declaring a slower process makes the budget looser; that choice must be tied to an external ecological hypothesis rather than selected to pass the gate.

4.4 Controls and edge cases

Table 1. Analytic controls defining the scope of the coherence claim.

Case	Result	Interpretation
All τ_i equal	$\eta = 1$	Common timestamp preserves the mode.
$\omega = 0$	$\eta = 1$	Static differences are unaffected by age.
One active channel	$\eta = 1$	Coherence alone does not imply completeness.
Two equal channels, half-cycle apart	$\eta = 0$	An average can erase the common signal.
Two equal channels, full-cycle apart	$\eta = 1$	Raw delay variance is sufficient, not necessary.

For two channels with weights w and $1 - w$ and delay difference $\Delta\tau$, Eq. (8) reduces to

$$\eta = 1 - 4w(1 - w) \sin^2\left(\frac{\omega\Delta\tau}{2}\right). \quad (13)$$

The worst cancellation occurs at balanced weights and an odd half-cycle. In contrast, an integer full-cycle difference is phase aligned. This periodicity is why Eq. (12) must never be represented as a necessary condition.

5 Verification

The exact program `verification.py` uses only the Python standard library and a fixed seed. It was written and executed before this manuscript. The captured, untruncated output is `verification_output.txt`.

The gate passed 21 of 21 checks. It included a hand-auditable three-channel case, 5,000 randomized cases with two to twelve channels, weights spanning seven orders of magnitude, delays on both sides of the reference time, and frequencies spanning more than four orders of magnitude. The maximum pairwise-identity error was 3.531×10^{-14} . The largest computed excess above the variance bound was 1.178×10^{-16} , consistent with floating-point roundoff.

An independent time-domain calculation sampled delayed cosines at 256, 1,024, 4,096, and 16,384 points and reproduced the phasor energy to below 2.5×10^{-15} . The small-delay experiment observed convergence orders 1.9985, 1.9996, and 1.9999. A four-frequency mixture passed the spectral bound, and a grid of 1,001 frequencies passed the operational $q = 0.90$ budget.

These checks reduce implementation and algebra risk but are not the formal proof. Formal support comes from the separate audited Lean artifact. The numeric suite does not validate the common-mode assumption, establish novelty, or demonstrate field utility. The small-delay convergence and spectral-mixture experiments are numeric checks of claims not separately stated as Lean theorems.

6 Empirical verification plan

A field test should begin with a named common-mode hypothesis, not with the composite itself. For example, a monitoring program might preregister that several normalized indicators respond with positive sensitivity to a seasonal productivity mode over a specified frequency band. Training years would be used to estimate signs, gains, delay distributions, and Ω . Weights, q , and interpolation rules would then be frozen.

On held-out years, the program would compare: (i) the original asynchronous composite; (ii) a common-time reconstruction using only training-fitted state-space or interpolation models; and (iii) a simple maximum-age baseline. The central test is whether snapshots inside the variance budget achieve coherence at least q against an independent common-mode reference, with delay uncertainty propagated. Snapshots outside the budget must remain in the test set as predicted failures.

The model should be rejected for the intended application if the channels lack the preregistered common mode, if estimated signs differ, if alignment does not improve held-out coherence, if the bound fails after delay uncertainty is included, or if the simpler baseline predicts failures equally well. Genuine ecological phase differences should be modeled as biology, not “corrected” away to make a score smoother.

7 Discussion

The main contribution is not the observation that timestamps matter. It is a compact contract connecting three quantities a monitoring product can declare: the frequency band it claims to represent, the coherence it promises, and

the weighted dispersion of the timestamps it actually uses. The pairwise identity also supports diagnosis: channels with large $w_i w_j$ and large phase separation dominate predicted loss, so remediation can target the specific stale pair rather than refresh every stream.

Run 118 is distinct from the prior D-Score lineage. Measurement closure asks whether a value is statistically resolvable. Monitoring water-filling asks where finite observation power should go. Revisit scheduling asks when each site should be observed again. Manipulation-proof weighting asks how strategic elasticity should affect weights. Snapshot alignment asks whether the chosen values, weights, and ages can represent one common-time state. A mature system would need all of these gates; passing one does not imply another.

The result also supplies a useful negative lesson. “Fresh enough” cannot be defined by a universal maximum age. A one-month mismatch may be negligible for a decadal mode and destructive for a seasonal or disturbance mode. Conversely, large age differences can phase-wrap for a single exact sinusoid, although such coincidence is fragile under broadband dynamics. The declared ecological bandwidth, not the calendar alone, is the essential comparison scale.

8 Limitations

The analysis is intentionally narrow. A composite may be temporally coherent yet biased, gamed, underpowered, spatially mismatched, or unrelated to ecological value. Nonnegative weights exclude validated anticorrelated channels. A single common mode excludes component-specific dynamics. Harmonic analysis is local and does not by itself handle regime shifts, hysteresis, irregular missingness, aliasing, nonlinear response, or uncertain clocks. The bound becomes negative and uninformative when $\Omega^2 \text{Var}_w(\tau) > 1$.

Most importantly, real ecological asynchrony is not automatically a defect. Phenological divergence can encode interactions, adaptation, or stability. The proposed gate applies only when channels are contractually intended to track the same local mode. No external dataset was analyzed, no field performance is claimed, and the targeted literature search cannot establish novelty or priority.

9 Reproducibility and declarations

Reproducibility. Run `python3 verification.py` with Python 3.11 or later; no third-party package or external data is required. The seed, tolerance, hypotheses, assumptions, and falsifiers are printed in the captured output. The manuscript compiles from `paper.tex` with Tectonic 0.17.0.

Data and code availability. All code and synthetic parameters used in this generated exploration are contained in the immutable Run 118 package. No external ecological data were generated or analyzed.

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Conflicts of interest. Justin D. Hart is founder of Viridis LLC, which may develop biodiversity measurement and assurance products. The separate business note reports zero signed revenue and no named customer validation attributable to this result.

Accountable authorship. Justin D. Hart is the accountable human author and retains responsibility for scientific claims, corrections, and any future submission. No AI system is represented as a journal author.

Substantive AI-use disclosure. Under Justin D. Hart’s direction, OpenAI Codex searched the local Viridis corpus and current literature, proposed and preregistered the hypothesis, wrote and executed the verification code, drafted the manuscript, assembled the package, and reconciled this final version line by line against the landed formal artifact. Historical Claude-authored Viridis artifacts were read for context with attribution preserved. Aristotle served as the automated theorem-proving system for the six frozen Lean targets; Codex then independently rebuilt and audited the returned artifact for statement freeze, proof holes, forbidden constructs, allowed axioms, non-vacuity, and significance. AI systems are tools and contributors, not authors or creators. No external peer review has occurred.

Stage and route. This version is the exact `FINAL_RECONCILED` manuscript bound to the local package. The authoritative ledger row is `null`: no exact Run 118 identity occurs in the root ledger. The advisory route is working corpus. The result is a domain-applying biodiversity-measurement theorem and is not admitted to the spine (`spine_admitted=false`). These local states confer no ledger approval, publication authorization, peer acceptance, adoption, or revenue.

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